PROJECT REPORT

Title: India's Agricultural Crop Production Analysis

INTRODUCTION

1.1 Project Overview

- India's
-
- Agricultural crop production analysis project
- Need for agricultural data visualization and analysis
- Excel +Tableau + MySQL + Flask overview

1.2 Purpose

- Why this project is developed
- Benefits for farmers, researchers, policymakers, decision-makers

---

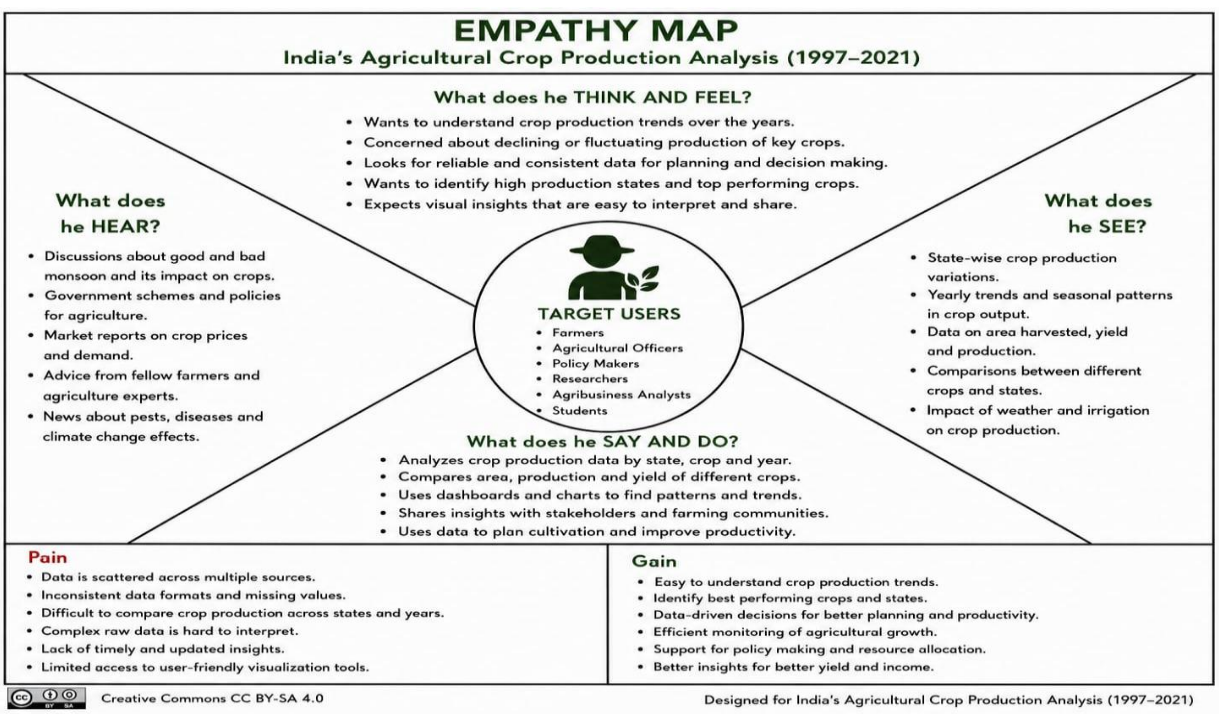
2.IDEATION PHASE

Problem Statement

India's agricultural sector plays a vital role in the country's economy, making it essential to analyze crop production trends across different states, seasons, and years. This project focuses on transforming agricultural data into meaningful insights by analyzing crop production, cultivated area, yield, and seasonal patterns from **1997-2021**

Using **Tableau**, the data is presented through interactive dashboards and story points, enabling users to explore state-wise production, crop-wise performance, seasonal variations, and overall agricultural trends. These visualizations help stakeholders make informed, data-driven decisions to improve agricultural planning, productivity, and resource management**.**

1.3 Empathy Map Canvas



Brainstroming:

Agricultural sector generates a large amount of crop production data across different states, districts, seasons, and crop types. The project aims to develop an interactive visualization dashboard that transforms raw agricultural data into meaningful insights, enabling users to analyze crop production trends, seasonal patterns, state-wise performance, and production distribution for better decision-making.

Brainstorming Ideas:

Data Analysis

- Agricultural crop Production dataset collection
- SQL database creation
- Data preprocessing and cleaning
- India's agricultural crop production analysis

Visualization

- Interactive Tableau dashboard
- State-wise crop production map
- Crop-wise yield and production comparison
- Seasonal and yearly production trends
- Dynamic filters and dashboard actions

Agricultural Insights

- State-wise crop production performance
- Crop yield comparison
- Area harvested and production analysis
- Seasonal crop production trends
- Top-performing crops and states

Web Integration

- Tableau Public publishing
- Dashboard embedding
- User-friendly web interface

Idea Prioritization:

The selected idea is the development of an Indian Agricultural Crop Analytics Platform by integrating MySQL database management and Tableau visualization.

The solution provides interactive insights into crop production, yield, cultivated area, seasonal trends, and state-wise agricultural performance, enabling better analysis and data driven decision-making.

---

2. REQUIREMENT ANALYSIS

2.1 Customer Journey Map

Need Information about Agricultural Crop Production

↓

Search for Crop Production Data Online

↓

Visit Agricultural Crop Production Analytics Web Application

↓

Open Tableau Dashboard / Story

↓

Apply Filters & Explore Visualizations

↓

Compare States, Crops, Seasons, Production & Yield

↓

Analyze Insights

↓

Make Data-Driven Agricultural Decisions

↓

Share Insights & Recommendations

2.2 Solution Requirement

Functional Requirements

The Indian agricultural crop production analysis provides the following functionalities:

- Collect and process agricultural crop production datasets from multiple sources.
- Store agricultural data information using a MySQL database.
- Retrieve and analyze crop production data using SQL queries.
- Create interactive Tableau visualizations for better data understanding.
- Display state-wise, crop-wise, season-wise production, area, and yield analysis.
- Provide dynamic filters and dashboard actions for user interaction.
- Publish Tableau dashboards and stories through Tableau Public.
- Integrate Tableau visualizations with a web-based interface.

Non-Functional Requirements

- The system should provide accurate analytical results.
- Dashboard interaction should be smooth and responsive.
- Visualizations should be simple and easy to understand.
- The system should support future dataset expansion.
- Published dashboards should be accessible through online links.

2.3 Data Flow Diagram

Agricultural Dataset Sources

↓

Data Collection Process

↓

MySQL Database

↓

SQL Data Processing Tableau Visualization

↓

Flask Web Application

↓

End Users

2.3 Technology Stack

Database Technology

MySQL Database

MySQL is used for storing and managing agricultural crop production datasets. It helps organize structured data and execute SQL queries for analysis.

Tools:

• MySQL Server 8.0

• MySQL Workbench

Visualization Technology

Tableau

Tableau is used for creating interactive dashboards and stories from processed agricultural datasets. Features Used:

• Worksheets

• Dashboards

• Stories

• Filters

• Maps

• Tableau Public Publishing

Web Technology

Flask Framework

Flask is used to create a web interface for displaying embedded Tableau dashboards and stories.

Technologies:

• Python Flask

• HTML

• CSS

Programming and Query Language

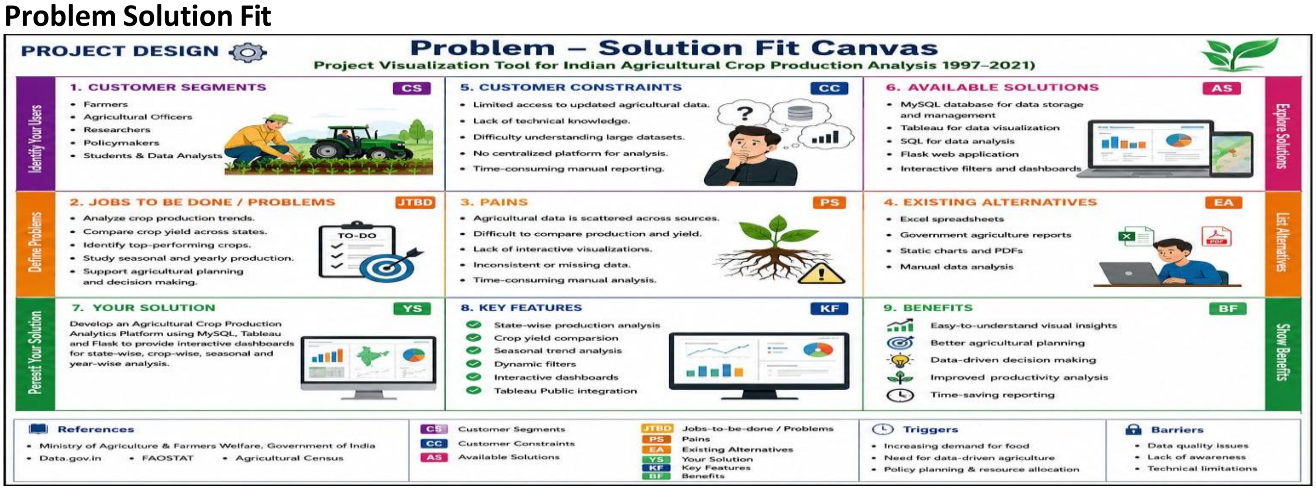
• SQL - Database operations and analysis

• Python - Web application development

---

3. PROJECT DESIGN

3.1 Problem Solution Fit



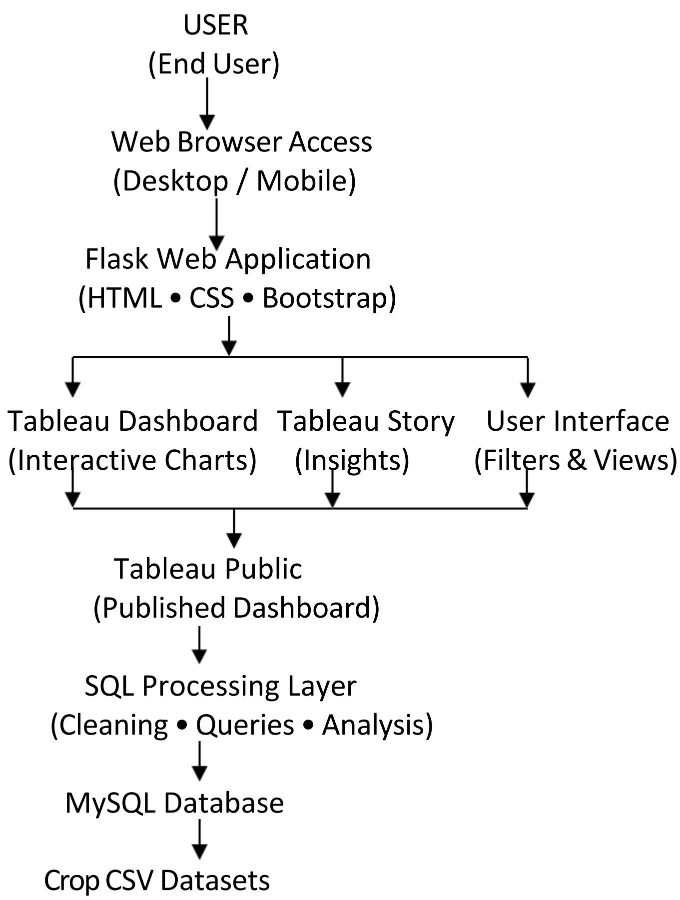
The Problem Solution Fit explains how the Indian Agricultural Crop Production Analytics Platform addresses the challenges faced by users in analyzing agricultural production data.
The solution combines agricultural crop datasets and provides interactive Tableau dashboards to analyze crop production, cultivated area, yield, seasonal trends, and state-wise performance.

3.2 Proposed Solution

The proposed solution is an Indian Agricultural Crop Production Analytics Platform that integrates MySQL database management, Tableau visualization, and Flask web technology.
The system collects agricultural crop production datasets from different sources and stores structured data in a MySQL database. SQL operations are performed for data processing, cleaning, and analysis.
The processed data is connected with Tableau to create interactive worksheets, dashboards, and stories. Users can analyze state-wise crop production, crop yield, cultivated area, seasonal trends, year-wise production, and top-performing crops and states. The final Tableau dashboard and story are published through Tableau Public and integrated into a Flask web Solution

---

Architecture



The solution architecture represents the complete workflow of the Electric Vehicle Analytics Platform. The system begins with EV CSV datasets which are processed and stored inside the MySQL database. SQL operations are performed for data preparation and analysis

The processed data is connected with Tableau Desktop to create interactive dashboards, charts, maps, KPIs, and stories.

The dashboards are published through Tableau Public Server and embedded into a Flask web application using HTML, CSS, and JavaScript.

End users such as agricultural customers and analysts can access the web application to explore agricultural crop insights interactively.

4. PROJECT PLANNING & SCHEDULING

Project planning defines the complete development process of the India’s agricultural crop production Platform. The project activities were divided into different sprint phases to complete development in an organized manner.

The planning process includes dataset preparation, database creation, visualization development, dashboard implementation, Tableau publishing, web integration, testing, and documentation.

Sprint	Activities	Duration
Sprint 1	Dataset Collection, Data Cleaning, MySQL Database Creation SQL Processing	24 Jun 2026 - 26 Jun 2026
Sprint 2	Tableau Worksheets, Dashboard Creation, Filters and Action Implementation	27 Jun 2026 - 30 June 2026
Sprint 3	Tableau Story Creation and Tableau Public Publishing	01 Jul 2026 - 03 Jul 2026
Sprint 4	Flask Web Integration, Testing and Documentation	04 Jul 2026 - 05 Jul 2026

5. FUNCTIONAL AND PERFORMANCE TESTING

Functional and performance testing was performed to verify the working efficiency, accuracy, and responsiveness of the India’s agricultural crop production analysis.

Testing ensures that datasets are properly processed, Tableau dashboards work correctly, filters respond accurately, and users can interact with the published analytics platform without issues.

Data Rendering Testing

The agricultural datasets were successfully imported into MySQL database and connected with Tableau Desktop for visualization.

Datasets Tested:

- Crop Production Dataset
- State-wise Agricultural Production Dataset
- Crop Yield Dataset
- Area Harvest Dataset

The database connection and data rendering process were verified successfully.

Visualization Testing

All Tableau worksheets and visualizations were tested to ensure proper data representation.

Visualizations Tested:

- KPI Dashboard (Total Production, Yield, Crop Count
- State-Wise Agricultural Production Distribution
- Year-wise total agricultural production
- Comparison of Cultivated area by crop

- Production Distribution by Unit Type
- Leading states by Agricultural Production
- All visualizations generated expected analytical results.

Publishing and Web Integration Testing

The completed Tableau dashboard and story were published successfully using Tableau Public.

The Tableau Public embed code was integrated into the Flask web application, allowing users to access interactive agricultural analytics through a web interface.

The web page responsiveness and dashboard loading performance were successfully verified.

Testing Result:

The India's agricultural crop production Analytics Platform successfully passed functional and performance testing. The system provides accurate visualization results, interactive filtering, and smooth dashboard accessibility.

---

6. RESULTS

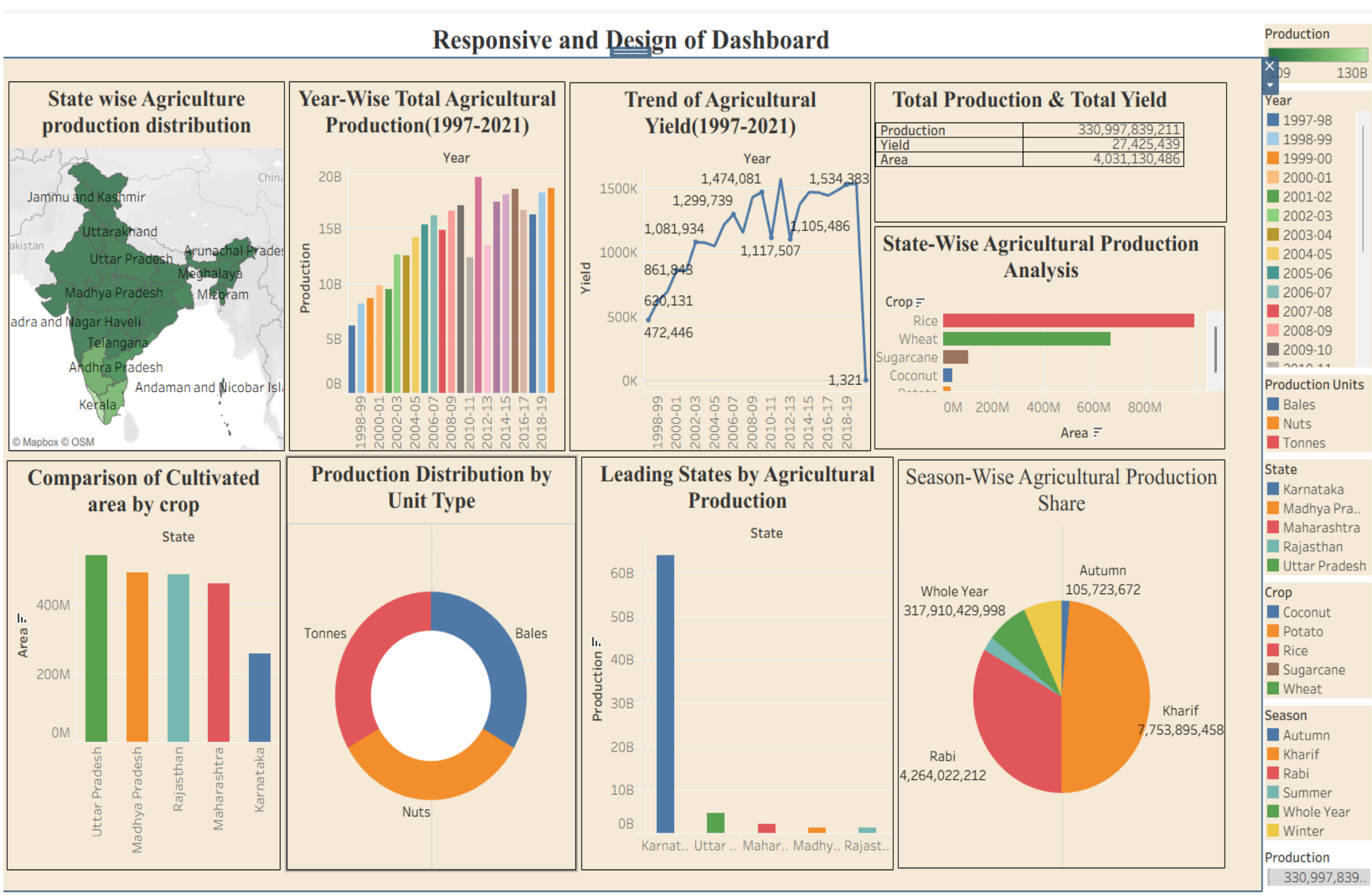
6.1 Output Screenshots

The India's agricultural crop production Analytics Platform was successfully developed by integrating MySQL database, Tableau visualization, and Flask web application.

The final system provides interactive dashboards and stories that allow users to analyze agricultural.

---

1. Tableau Dashboard Output

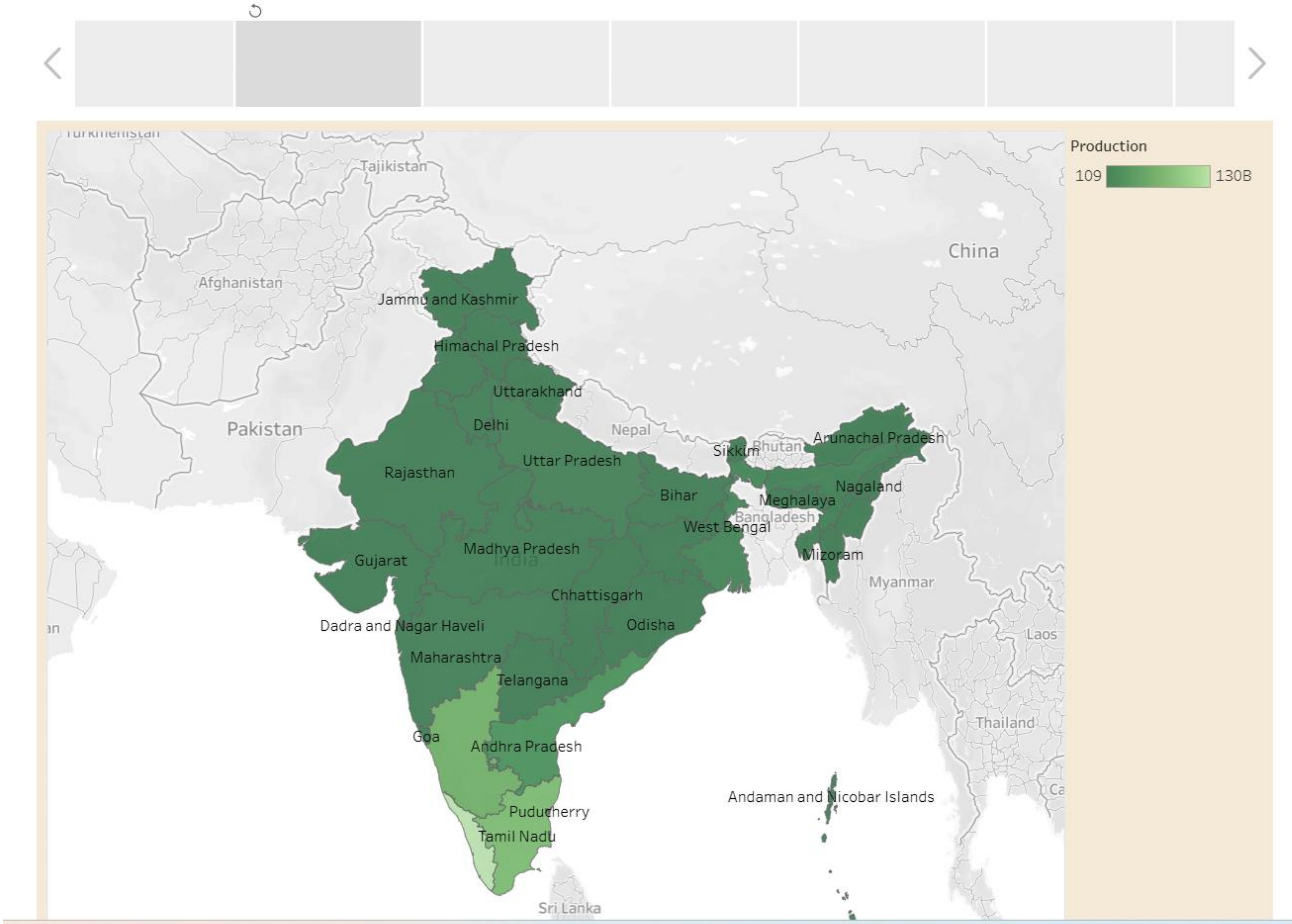


The interactive Tableau dashboard provides a comprehensive analysis of India's Agricultural Crop Production by presenting key insights such as crop production, yield, area harvested, seasonal distribution, and state-wise agricultural performance. The dashboard enables users to compare different crops, identify top-producing states, analyze production trends, and evaluate seasonal variations through interactive charts and maps..

Users can interact with dashboard filters to explore specific agricultural information and gain meaningful insights for data driven decision-making.

---

2. Tableau Story Output



The Tableau story presents India’s agricultural production analysis in a sequential storytelling format. It helps users understand important agricultural trends and insights step by step.

---

3. Web Application Output

The Flask web application provides a user-friendly interface where Tableau dashboards and stories are embedded.

Users can access agricultural analytics directly through the web application without opening Tableau separately.

7. ADVANTAGES & DISADVANTAGES

Advantages

The advantages of the India's agricultural Production Analysis Platform are:

- Provides interactive visualization of agricultural data through Tableau dashboards and stories.
- Integrates multiple agricultural datasets into a centralized analytics platform for comprehensive analysis.
- Enables users to compare crop production, Yield, and area harvested across different states and years.
- Offers geographical visualization of agricultural charging station distribution using interactive maps.
- Dynamic filters allow users to explore agricultural production using interactive map.
- SQL-based data analysis improves data retrieval, filtering, aggregation, and reporting efficiency.
- Flask web application provides a simple and user-friendly interface for accessing dashboards and stories.

Disadvantages

The limitations of the India's agricultural crop production analysis Platform are:

- The analysis depends on the accuracy and availability of the collected agricultural datasets.
- Real-time agricultural data integration is not implemented in the current system.
- Tableau Public dashboards require an internet connection for online access.
- The system focuses on visualization and does not include AI or predictive analytics.
- New datasets must be manually cleaned and imported into the database.

8. CONCLUSION

The India's Agricultural Crop Production Analysis Platform was successfully developed to analyze and visualize agricultural crop production data using MySQL and Tableau.

The project integrated multiple agricultural datasets containing crop production, yield, area harvested, seasonal information, and state-wise agricultural statistics. The datasets were cleaned, processed using MySQL database operations, and transformed into meaningful visual insights through Tableau dashboards.

The interactive dashboard enables users to analyze crop production trends, compare agricultural performance across different states, evaluate seasonal variations, and identify major crop-producing regions using dynamic filters and visualizations. Overall, the project provides a simple, efficient, and user-friendly platform for supporting agricultural analysis and data-driven decision-making.

9. FUTURE SCOPE

The future enhancements of the India's agricultural crop production analysis Platform include:

- Integration of real-time agricultural datasets for continuous updates.
- Implementation of machine learning models to predict crop production and Yield.
- Development of an crop recommendation system based on user requirements.
- Addition of weather forecasting and rainfall analysis to improve agricultural planning.
- Integration of satellite and remote sensing data for crop monitoring

• Inclusion of market price analysis to help farmers make informed crop-selling decisions.

• Expansion of the platform to support district-wise and village wise agricultural analysis.

• Integration with government agricultural schemes and advisory services for better decision-making.

10. APPENDIX

Source Code

The complete source code contains:

• Flask application files
• HTML and CSS user interface
• Tableau integration code
• SQL database scripts
• Project datasets

GitHub Repository Link:

Dataset Link

Datasets used in this project:
1.Area analysis
2.Crop production analysis
3.Season Analysis
4.Yield Analysis

GitHub & Project Demo Link
https://drive.google.com/file/d/1rHB2zq_WL2cq1TjosEmkRfc6sCGa1EuW/view?usp=drivesdk
GitHub Repository:

https://github.com/mounikamandula23-bit/India-s-Agricultural-Crop-Production-dashboard

Tableau Dashboard Public Link:

https://public.tableau.com/app/profile/mounika.mandula/viz/ResponsiveandDesignofDashboard/ResponsiveandDesignofDashboard

Tableau Story Public Link:

https://public.tableau.com/app/profile/mounika.mandula/viz/Year-WiseTotalAgriculturalProduction1997-2021/AgriculturalProductionOverview?publish=yes